



LightCOM

Lighthouse Consortium on
Obesity Management

Study name: **Lighthouse Consortium on Obesity Management**
Short name: **LightCOM**

SOP title: **Standard Operating Procedure for Venepuncture**

SOP short title: **SOP on Venepuncture**

SOP Version: **1** Date: **09.04.2024**

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1. PURPOSE

The purpose of this Standard Operating Procedure (SOP) is to describe the standard procedures to be followed when obtaining a blood sample using a needle and holder or a winged infusion device with participants during the LightCOM studies which includes LightBAR, LightWAY and LightCARE.

2. INTRODUCTION

Venepuncture is a commonly performed procedure to obtain a blood sample from patients, usually from the superficial veins in the upper extremities ie: the median veins in the antecubital fossa (basilic, median cubital and cephalic veins) because of their size and ease of access.

3. SCOPE

This SOP applies to clinical research carried out under the LightCOM research studies which incorporates the individual studies of LightBAR, LightWAY and LightCARE. The sponsor is Department of Medicine, Copenhagen University Hospital Amager and Hvidovre, Copenhagen, Denmark.

4. RESPONSIBILITIES

4.1 Chief Investigator/Principle Investigator or delegate

Will be responsible for ensuring the staff member assigned for obtaining blood samples using the venepuncture method is suitably qualified to undertake the procedure and has received adequate and up to date training.

5. SPECIFIC PROCEDURE

Check the participant's identity by asking name, D.O.B and by cross-checking with the patient study ID

Check the participant understands this is a fasting blood test and has fasted from 10PM the night before.

- Thoroughly cleanse your own hands with clinically appropriate hand sanitiser.
- Follow local policy on use of gloves and apron during the procedure.

Ask the participant to sit or lie in a comfortable position.

Inform the participant of the potential risks of the venepuncture procedure i.e.: bruising, haematoma and pain.

- Examine both arms and inspect for any signs of bruising, infection or phlebitis. Avoid any veins that are fibrosed, inflamed or fragile.
- Check if they had had previous problems with the venepuncture procedure.

Check if the participant has had surgery on or under the arm or any preference as to which arm to use, then place a pillow under the arm to provide support for venous access.

- Locate a vein to use by either:
 - Visual inspection - look for an obvious full vein
 - Palpitation – the vein should be soft, bouncy and refill on palpation

- If local policy dictates, clean the venepuncture area with an alcohol-based solution
- Apply a tourniquet about 7-8 cm above the intended venepuncture site. Adjust the tourniquet so that it is tight enough to impede venous return but not so tight the radial pulse cannot be palpated.
- Do not leave the tourniquet in place for longer than 1 minute as this can lead to inaccurate blood results.
- If you have difficulty with venous access, consider:
 - Remove the tourniquet, place a heat compress to the vein area
 - Remove the tourniquet, ask participant to lower their arm below the level of their heart to encourage venous fill.
 - Leave the tourniquet in position and gently palpate, lightly tap or stroke the vein to encourage the release of histamine and vasodilation.
 - Remove the tourniquet and ask the participant to clench and open their fist. (only use if other methods fail as clenching fist can affect blood plasma)
- Use one of the following procedures:

1. Venepuncture using needle and holder:

- Once venous access is established – stabilise the area using your thumb (make sure you use your hand not engaged in the needle and holder) to stretch the skin downwards.
- Unsheathe the needle and holding the plastic needle holder between thumb and index finger, position the needle over the vein.
- Smoothly insert the needle, bevel up, into the vein at an angle of between 10° and 30° depending on the depth and size of the vein. Advance the needle until it is securely in the vein.
- Whilst keeping the plastic holder and needle steady, using your free hand connect the vacuumed blood collection tube into the plastic holder. Gently push the collection tube to connect with the holder whilst ensuring you do not advance the needle further into the participant's vein.
- Once the collection tube is in the correct position in the holder, the vacuum in the tube will draw blood into it. Fill the tube to the required level following the manufacturer's guidance.
- Once the collection tube has enough blood as required, remove the tube from the holder whilst maintaining the needle position in the vein. Attach other collection tubes as required for the blood samples required (see appendix 1 for blood collection tubes).
- On the last collection tube, release the tourniquet. When the tube is filled to the required amount, remove from the needle as in previous collection tubes whilst maintaining the needle in position.
- Apply a gauze swab over the needle entry area and withdraw the needle. Ask the participant to apply pressure to the needle entry site by holding the swab in place.
- Follow the specific manufacturer's instructions in covering the needle i.e. action the safety cover by pushing the holder towards the needle then dispose of the needle and holder immediately into a sharps container.

2. Venepuncture using a winged infusion device:

- Attach the needle to the winged infusion device following specific manufacturer's instructions. Remove the plastic cover from the needle and fold the wings upwards and inwards.
- Once venous access is established, anchor the vein using your thumb (make sure you use your hand not engaged in the needle and holder) to stretch the skin downwards.
- Smoothly insert the needle, bevel up, into the vein at an angle of between 10° and 30° depending on the depth and size of the vein. Advance the needle until it is securely in the vein.
- You should see a small flashback of blood into the needle's chamber and a feel a decrease in resistance as the needle enters the vein. After this flashback of blood, advance the needle 1-2mm further into the vein to ensure it is within the lumen.
- Once the needle is securely in place in the vein, release the wings and insert a vacuumed blood collection tube.
- Once the collection tube is in the correct position in the holder, the vacuum in the tube will draw blood into it. Fill the tube to the required level following the manufacturer's guidance.
- Once the collection tube has enough blood as required, remove the tube from the holder whilst maintaining the needle position in the vein. Attach other collection tubes as required for the blood samples required (see Appendix 1).
- On the last collection tube, release the tourniquet. When the tube is filled to the required amount, remove from the needle as in previous collection tubes whilst maintaining the needle in position.
- Follow the manufacturer's advice on removal of the needle if it has a needle safety device, and activate this before removing the needle.
- Apply a gauze swab over the needle entry area (withdraw the needle if the above step isn't followed due to manufacturer's advice). Ask the patient to apply pressure to the needle entry site by holding the swab in place.
- Dispose of the needle and holder immediately into a sharp's container.

After both procedures:

- Check the participant's puncture site for any bleed and use an adhesive plaster (or additional cover if patient is allergic to adhesive plasters). Ensure the participant is comfortable.
- Gently invert the collection tubes to mix the samples taken with the additives – do not shake the collection tubes.
- Complete the Blood Request form with the patient details and tests required.
- Discard all other Clinical waste in appropriate bins.

6. FORMS/TEMPLATES TO BE USED

UK:

- For samples collected for 'immediate analysis', use local NHS trust specified Blood Request Forms, and label in the usual way.
- For samples collected for 'bioresource', attached the supplied barcodes and enter barcodes, time and date of sample in the eCRF. You will need to enter the barcode in twice to ensure accurate data entered

For DK see "Håndteringskema for biologisk materiale" for LigthBAR or LightCARE and WAY.

Record results on Study Database as per study protocol.

7. INTERNAL AND EXTERNAL REFERENCES

7.1 Internal References

Design and Development of SOP – CTRG Template SOP V3.2

7.1 External References

Venepuncture-Needle-and-Holder-P1-P5 @2023 Clinical Skills

8. HISTORICAL RECORD OF SOP

Rev# / Date	Written/Revised By	Revision Description	Approval
0. Written	Odette Dawkins		
1. Edited	Sabina Wielsøe	Merge of 1. Standard Operating Procedure for Venepuncture using a winged infusion device and 2. Procedure for Venepuncture using needle and holder	
2.			
3.			
4.			
5.			
6.			
Removed / replaced:			

10. APPENDIX 1

Samples for routine analysis

- 3.5 mL SST serum tube (gold top)
- 3 mL lithium heparin PST tube (green top)
- 4 mL EDTA tube (purple top)
- 2 mL fluoride oxalate (grey top)

Samples for the Biobank

- 9 mL sodium citrate tube (blue top)
- 8.5 mL SST serum tube (gold top)
- 10 mL lithium heparin tube (green top)
- 10 mL EDTA tube (purple top)